VIRTUAL CLOUD COMPUTING FUNDAMENTALS

Cloud Computing Internship – CodeOrbit Tech

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1. INTRODUCTION

Cloud computing provides computing resources such as servers, storage, databases, networking, and software through the internet. Users can access resources when required instead of maintaining physical systems.

It is widely used because it provides flexible and scalable resources. Major providers include Amazon Web Services (AWS), Microsoft Azure, and Google Cloud.

2. WHAT IS CLOUD COMPUTING?

Cloud computing is the delivery of computing services over the internet, including computing power, storage, databases, networking, and software.

For example, users can store files using cloud storage services, while organizations can create virtual servers without purchasing physical servers. Resources can be increased or decreased according to requirements.

3. IAAS – INFRASTRUCTURE AS A SERVICE

Infrastructure as a Service (IaaS) provides virtualized computing infrastructure through the internet, including virtual machines, storage, and networking. Users can configure the operating system and applications.

Example: Amazon EC2 allows users to create and manage virtual servers in the cloud.

Benefits: Flexible infrastructure; scalable resources; no need to purchase physical servers; configurable virtual machines.

4. PAAS – PLATFORM AS A SERVICE

Platform as a Service (PaaS) provides a platform for developers to build, test, and deploy applications. The cloud provider manages the underlying infrastructure so developers can focus on application development.

Example: Microsoft Azure provides platform services for developing and deploying applications.

Benefits: Simplifies development, reduces infrastructure management, supports deployment, and helps developers focus on coding.

5. SAAS – SOFTWARE AS A SERVICE

Software as a Service (SaaS) provides software applications through the internet. Users can access software without installing and maintaining the complete application infrastructure.

Examples: Google Workspace and Microsoft 365.

Benefits: Easy internet access, no complex installation, access from different devices, and provider-managed maintenance.

6. MAJOR CLOUD SERVICE PROVIDERS

1. Amazon Web Services (AWS) – Provides computing, storage, databases, networking, and other cloud resources.

2. Microsoft Azure – Provides application development, virtual machines, storage, databases, and other cloud services.

3. Google Cloud – Provides computing, storage, databases, networking, and other cloud services.

7. COMPARISON OF AWS AND MICROSOFT AZURE

Feature	AWS	Microsoft Azure
Provider	Amazon	Microsoft
Compute Service	Amazon EC2	Azure Virtual Machines
Storage Service	Amazon S3	Azure Blob Storage
Database Service	Amazon RDS	Azure SQL Database
Main Use	Cloud infrastructure and applications	Cloud applications and Microsoft-oriented workloads

AWS and Microsoft Azure are both major cloud providers offering computing, storage, networking, and database services. The choice depends on the requirements of the organization or project.

8. EXAMPLES OF CLOUD COMPUTING

Cloud computing is used for online file storage, web hosting, application development, database hosting, backup and recovery, online collaboration, virtual machines, and cloud-based software applications.

9. ADVANTAGES OF CLOUD COMPUTING

- Scalability – Resources can be increased or decreased according to requirements.
- Flexibility – Cloud services can be accessed through the internet.
- Cost efficiency – Organizations can avoid large investments in physical infrastructure.
- Accessibility – Services and data can be accessed from different locations.
- Resource management – Cloud providers manage the underlying infrastructure.
- Faster deployment – Cloud resources can be created and configured quickly.

10. CONCLUSION

Cloud computing provides flexible and scalable computing resources through the internet. The three major service models are IaaS, PaaS, and SaaS. Major providers such as AWS, Microsoft Azure, and Google Cloud offer services for computing, storage, databases, networking, and application development.

Understanding cloud computing fundamentals provides a strong foundation for working with cloud platforms and services.

REFERENCES

Amazon Web Services – https://aws.amazon.com/

Microsoft Azure – https://azure.microsoft.com/

Google Cloud – https://cloud.google.com/